

From Points to Prints

Monthly Presentation (2)

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IGN

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Context 1

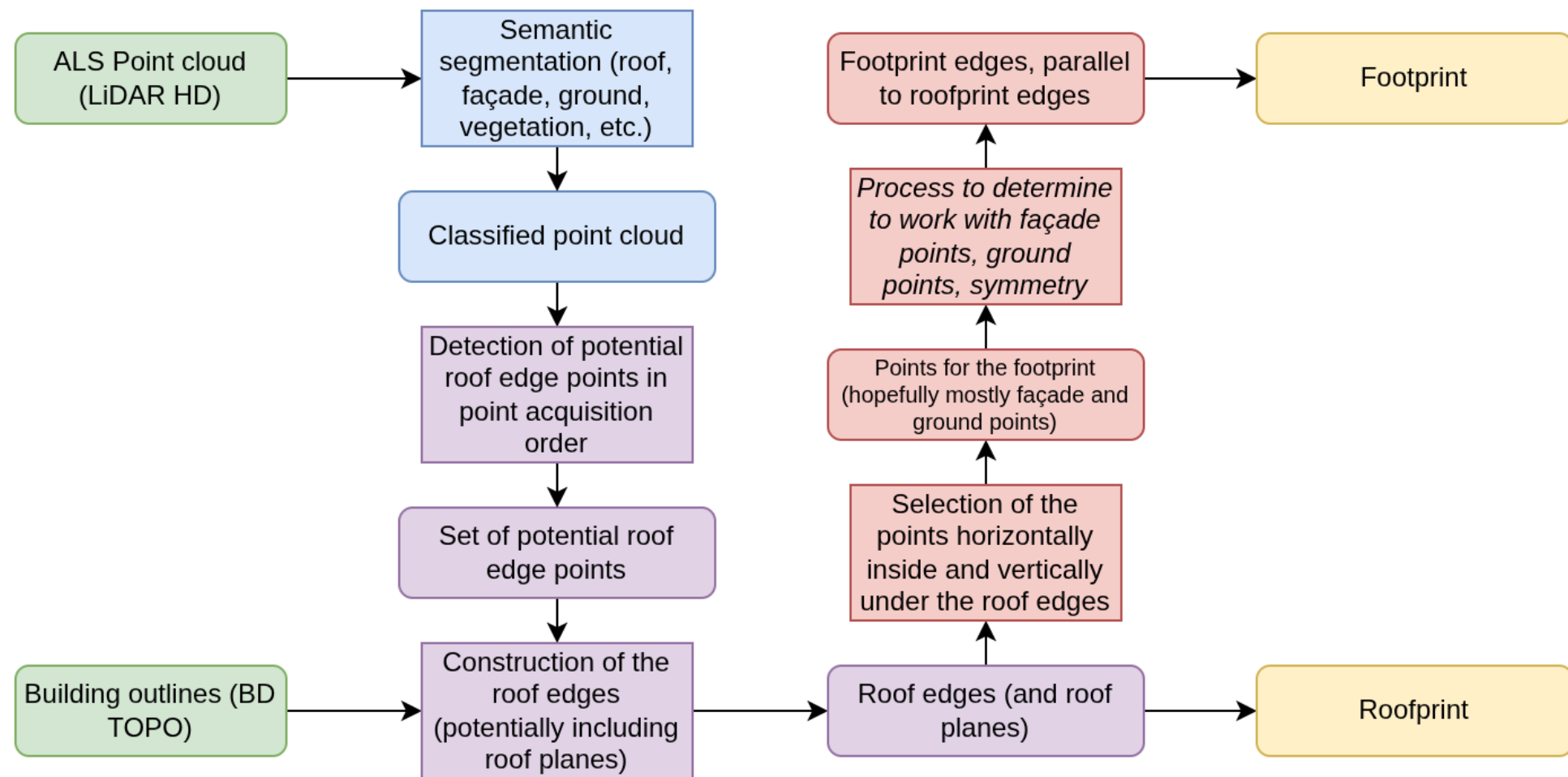
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Context

Context



Work done until now

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Work done until now

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Pulse

A single emission of the LiDAR sensor, which may result in one or more echoes (points) depending on the number of surfaces the pulse hits.

Scan line

A set of points collected during one rotation of the LiDAR scanner, i.e., one “scan” of the platform.

Flight strip

A continuous swath of LiDAR data collected along one pass of the aircraft over the ground, i.e., one “track” of the platform.

Work done until now
– Point cloud topology

Point cloud topology

LiDAR HD is distributed as a set of **cloud-optimized tiles** (1000m x 1000m) in the COPC format. This means that **flight strips are mixed together** in the same tile, and points are **spatially ordered** instead of being ordered by acquisition time.

Work done until now
– Point cloud topology

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Therefore, to extract the topology, we use:

- For flight strips: the **Point Source ID** field, which identifies the flight strip the point belongs to
- For scan lines:
 - The **GPS Time** field, which is a timestamp of the acquisition of the point, and can be used to sort points in acquisition order
 - The **Scan Direction Flag** field, which indicates the direction of the scan and alternates between 0 and 1 for consecutive scan lines
- For pulses: the **GPS Time** field, which is the same for all points of a pulse

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- For pulses: the **GPS Time** field, which is the same for all points of a pulse

The **Number of Returns** field is not reliable for pulses in the LiDAR HD dataset, as it is not updated when points are filtered out during the processing of the raw data.

Flight strips trajectories

I use code from Wu Teng to **retrieve the trajectory** of the scanning vehicle, using multi-echo pulses. The precision of the method increases with the number of multi-echo pulses, meaning that it is necessary to **rebuild the flight strips** (scattered between tiles) in order to get a good estimation of the trajectory.

Work done until now
– Flight strips trajectories

Flight strips trajectories

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This method allows to use the trajectory **without relying on its availability**, and therefore without adding new requirements on the input data.

Work done until now
– Flight strips trajectories

Identification of potential edge points

Points are identified by iterating over a flight strip in **acquisition order**. They are compared to the **lowest point in the neighbourhood**:

- The lowest point in the same pulse in case of a multi-echo pulse
- The lowest point in the previous and next pulses in case of a single-echo pulse

Work done until now

– Identification of potential edge points

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To account for the variation of angular difference between pulses in ellipsoidal LiDAR scanners, the **height difference** (Δh) between the points is divided by the **temporal difference** (Δt) between the pulses:

$$v = \frac{\Delta h}{\Delta t}$$

The temporal difference Δt is assumed to be a **good approximation of the angular difference**, as the scanning speed is constant.

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Work done until now

– Identification of potential edge points

Identification of potential edge points

Finally, this value is compared to a threshold $v_{\min}$, which is set to $v_{\min} = 2 \cdot 10^6$ with $(\Delta h)_{\min} = 2$ in metres and $(\Delta t)_{\min} = 10^{-6}$ in seconds. For multi-echo pulses, since $\Delta t = 0$, the height difference is directly compared to the threshold: $\Delta h > (\Delta h)_{\min}$.

Work done until now

– Identification of potential edge points

Identification of potential edge points

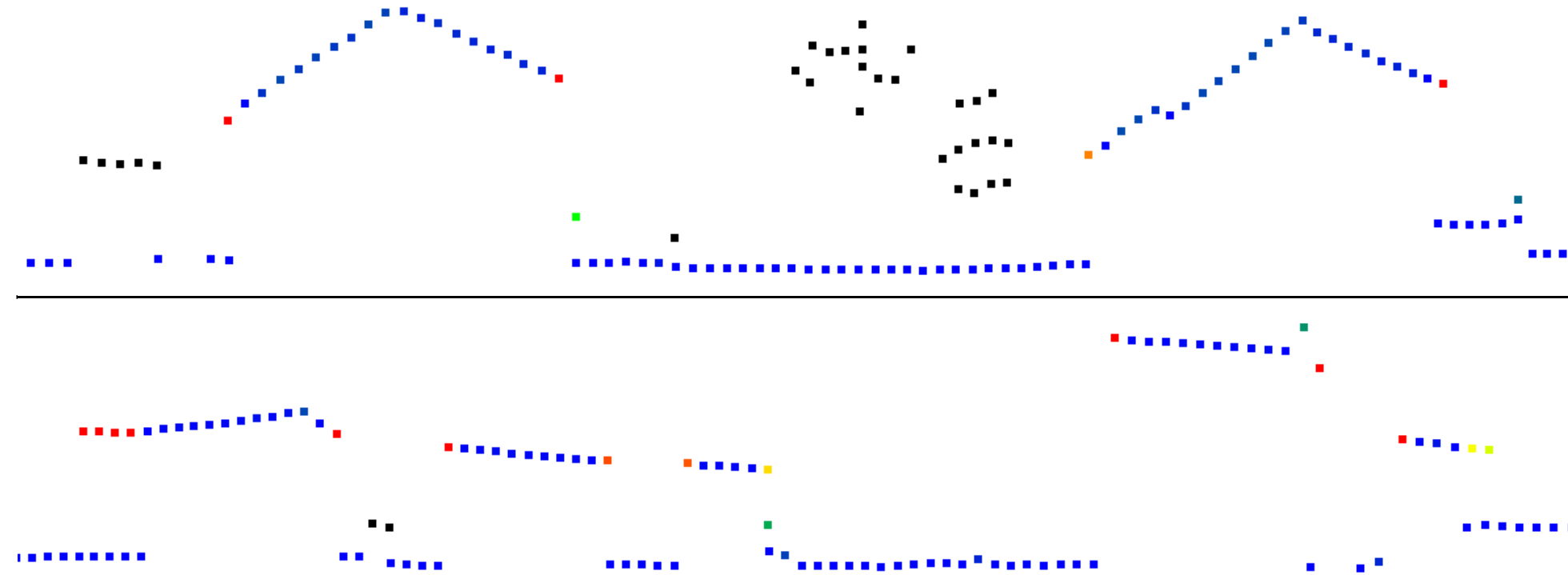


Figure 1: Two examples of the height differences obtained on a scan line. Black points are vegetation points, and for the other points the color represents the value of Δh , with blue-green-yellow-red from low to high values.

Work done until now

– Identification of potential edge points

- Points at the edges of what looks like building roofs have **high values of Δh**
- Other points get **low values of Δh**
- Points classified as **vegetation** would have gotten high values if not excluded
- These scan lines are actually **curved** when looked at from above, but this is not a problem as the **angle is very small** between consecutive pulses

Weighing of potential edge points

Potential edge points are weighted using:

1. Their **height** (normalized over the area of the building) to a factor $w_h \in [1.0, 3.0]$

Work done until now

– Weighing of potential edge points

1. The **height** is a **simple but effective** metric to give more weight to points a roof edge than to the points on the façade below it.
2. The **origin** gives more weight to real points that are **less likely to be imprecisely positioned**.
3. The **classification** gives more weight to points classified as building, which are **less likely to actually be something else such as vegetation**. This is important because many points are unclassified.

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1. Their **height** (normalized over the area of the building) to a factor $w_h \in [1.0, 3.0]$
2. Their **origin** to a factor $w_o \in \{0.1, 0.5, 1.0\}$:
 - Multi-echo: 1.0
 - Single echo with good estimation of position: 0.5
 - Single echo with less precise estimation of position: 0.1

Work done until now

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3. Their **classification** to a factor $w_c \in \{1.0, 2.0\}$:
 - Building: 2.0
 - Other: 1.0

Work done until now

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3. Their **classification** to a factor $w_c \in \{1.0, 2.0\}$:
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 - Other: 1.0

These values were picked arbitrarily and would need to be optimized. The final weight w is given by:

$$w = w_h \times w_o \times w_c$$

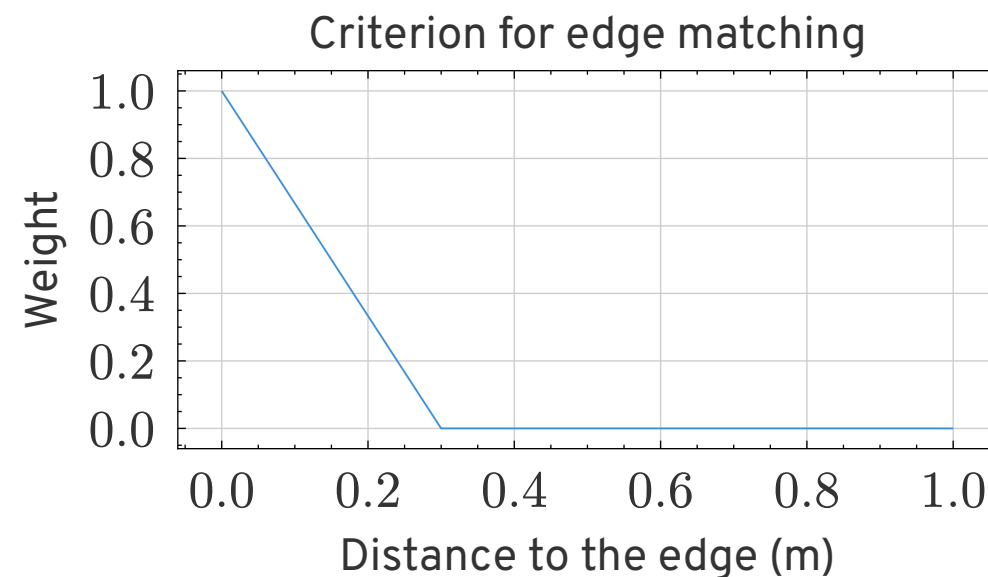
Work done until now

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Displacement of BD TOPO edges

The value of a point is not only determined by its weight, but also by its **distance to the edge**. Distances are computed **in 2D** after getting rid of the vertical component of the position.



Work done until now

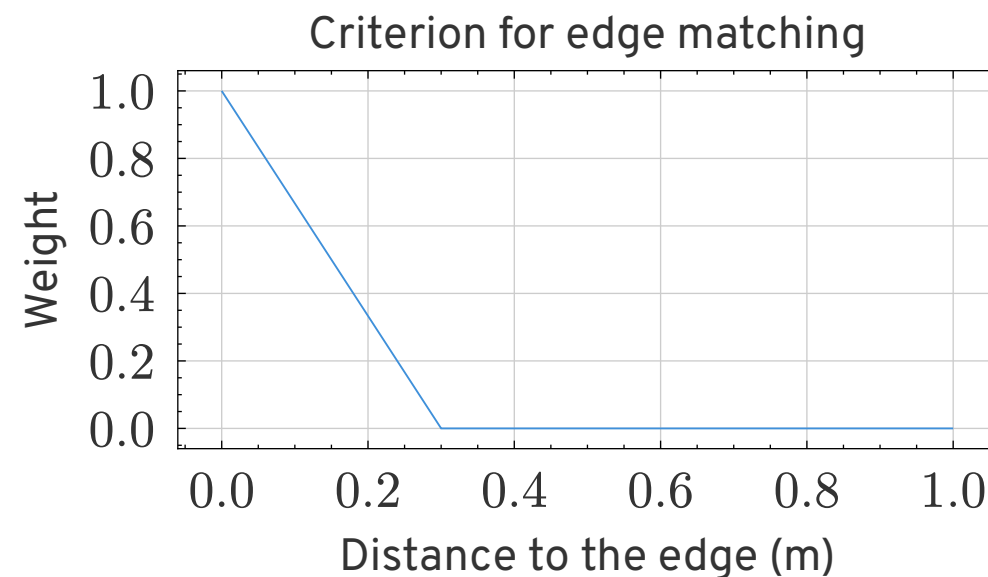
– Displacement of BD TOPO edges

- The decreasing weight based on distance to the edge allows to find the **best position** for the edge.
- The current threshold of **30 cm** was **picked arbitrarily** and would need to be optimized.
- Discarding points that **do not project on the segment** prevents matching to a longer edge that would be close but not actually the right one.

Displacement of BD TOPO edges

The value of a point is not only determined by its weight, but also by its **distance to the edge**. Distances are computed **in 2D** after getting rid of the vertical component of the position.

This distance is evaluated by projecting the point onto the edge, keeping only the points that project on the segment.



Work done until now

– Displacement of BD TOPO edges

- The decreasing weight based on distance to the edge allows to find the **best position** for the edge.
- The current threshold of **30 cm** was **picked arbitrarily** and would need to be optimized.
- Discarding points that **do not project on the segment** prevents matching to a longer edge that would be close but not actually the right one.

Displacement of BD TOPO edges

The BD TOPO edges are translated perpendicularly to themselves, and the criterion is evaluated for each translation.

Work done until now

– Displacement of BD TOPO edges

Preliminary results

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Identification of potential edge points



Figure 3: Edge points in blue and LiDAR HD data in gray scale from low (black) to high (white) intensity.

Preliminary results

- Identification of potential edge points

Identification of potential edge points

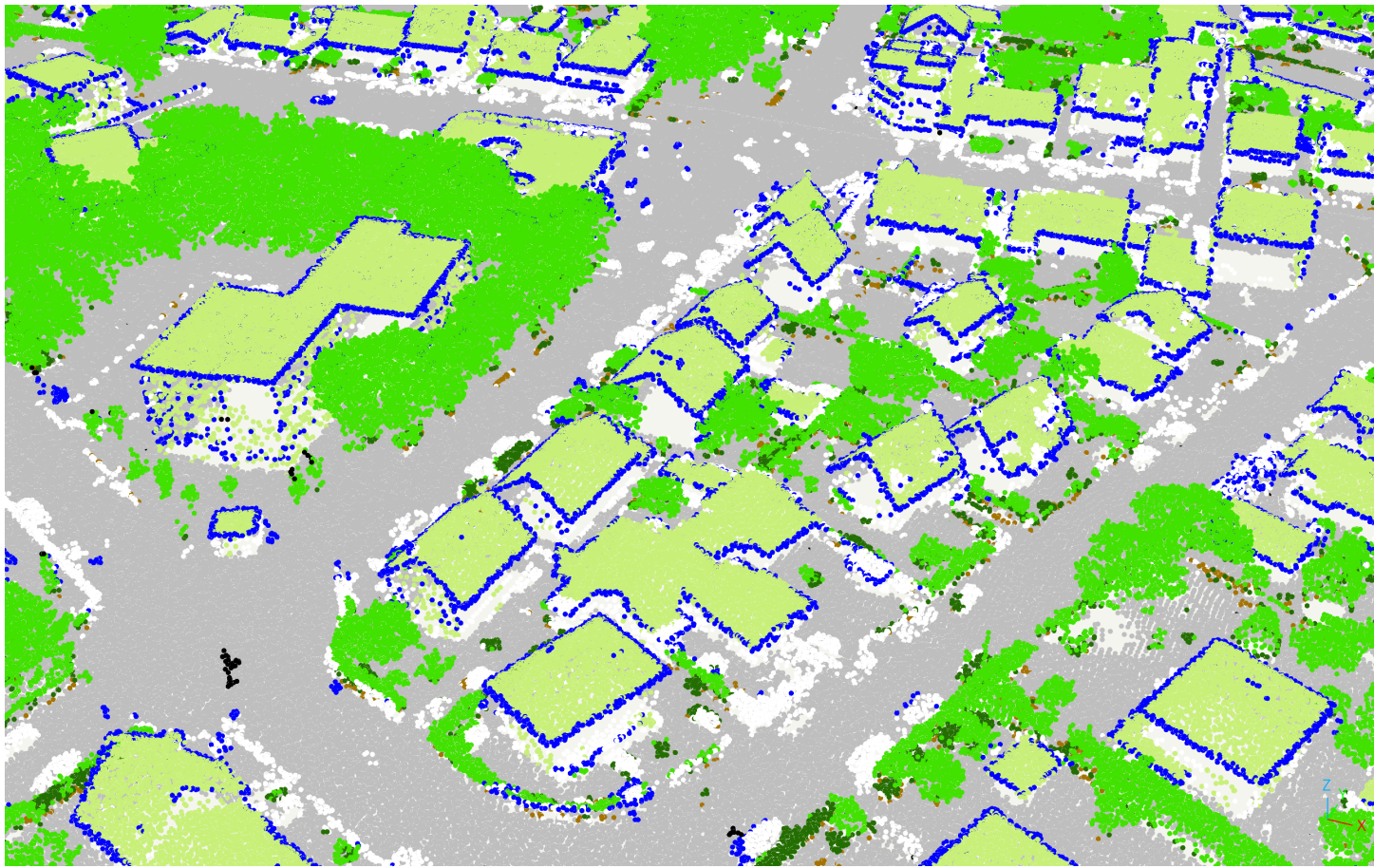


Figure 4: Edge points in blue and LiDAR HD data coloured by classification.

Preliminary results

- Identification of potential edge points

Identification of potential edge points

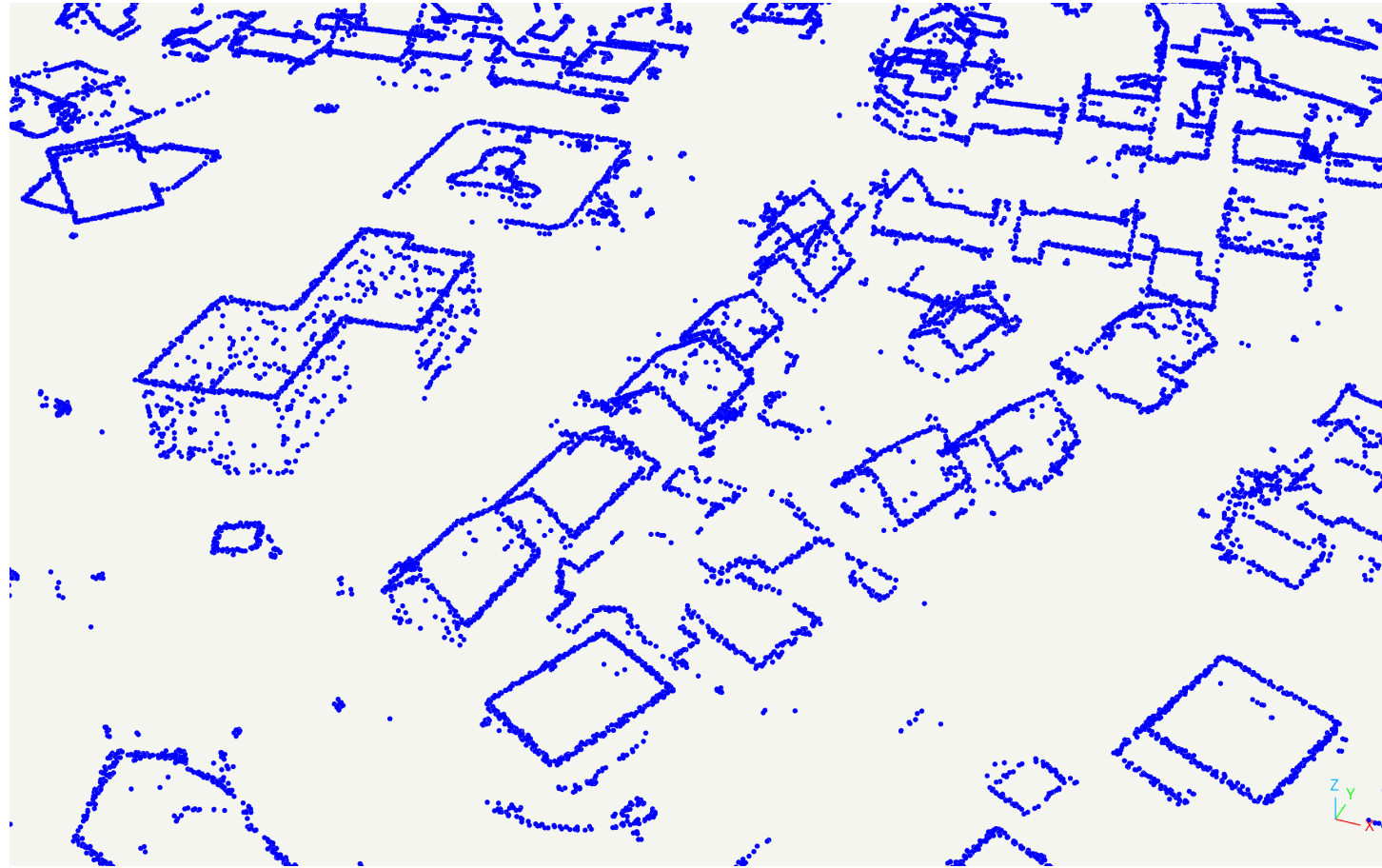


Figure 5: Edge points in blue without LiDAR HD data.

Preliminary results

- Identification of potential edge points

Identification of potential edge points

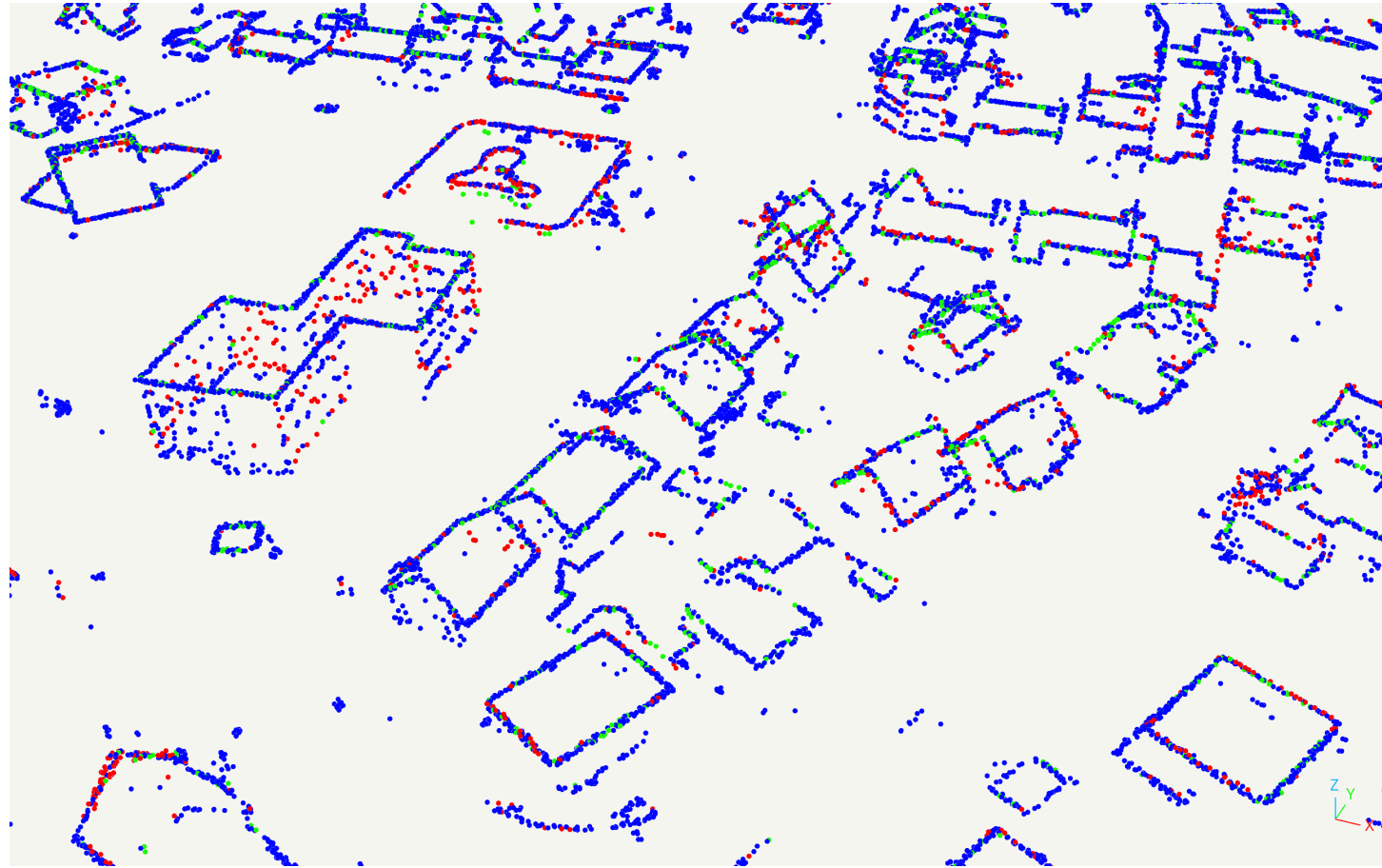


Figure 6: Edge points coloured per type (blue: real point, green: generated by translation, red: generated with trajectory) without LiDAR HD data.

Preliminary results

– Identification of potential edge points

Next objectives

Next objectives

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Thank You



Any questions?

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Next objectives

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